NET High-Yield Maths Formula and Trick Book

1. Number System and Algebra Basics

Indices and Surds

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}$$
$$(ab)^n = a^n b^n, \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}, \quad a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$$
$$a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m}, \quad \sqrt{ab} = \sqrt{a}\sqrt{b}$$
$$(a + b\sqrt{c})(a - b\sqrt{c}) = a^2 - b^2 c, \quad \frac{1}{a+\sqrt{b}} = \frac{a-\sqrt{b}}{a^2 - b}$$

Factorization and Absolute Value

$$a^2 - b^2 = (a-b)(a+b), \quad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$
$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$
$$|x-a| < b \Rightarrow a-b < x < a+b, \quad |x-a| > b \Rightarrow x < a-b \text{ or } x > a+b$$

Tricks: Use factorization before expansion. For $|x-a| < b$, write interval directly. Rationalize before matching options.

2. Logarithms and Exponents

$$\log_a xy = \log_a x + \log_a y, \quad \log_a \frac{x}{y} = \log_a x - \log_a y, \quad \log_a x^n = n \log_a x$$
$$\log_a a = 1, \quad \log_a 1 = 0, \quad a^{\log_a x} = x, \quad \log_a x = \frac{\log_b x}{\log_b a}$$
$$a^x = a^y \Rightarrow x = y, \quad a^x = b \Rightarrow x = \log_a b$$

Tricks: Check log domain: argument > 0, base > 0, base $\neq 1$. Convert to the same base.

3. Sets, Relations, Functions

$$A - B = A \cap B', \quad (A \cup B)' = A' \cap B', \quad (A \cap B)' = A' \cup B'$$
$$n(A \cup B) = n(A) + n(B) - n(A \cap B), \quad n(A - B) = n(A) - n(A \cap B)$$
$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$
$$R \subseteq A \times B, \quad (a,a) \in R \text{ reflexive}, \quad (a,b) \in R \Rightarrow (b,a) \in R \text{ symmetric}$$
$$(a,b), (b,c) \in R \Rightarrow (a,c) \in R \text{ transitive}$$
$$(g \circ f)(x) = g(f(x)), \quad (f \circ g)(x) = f(g(x)), \quad g \circ f \neq f \circ g \text{ usually}$$
$$y = f(x) \Rightarrow \text{solve for } x \Rightarrow f^{-1}(x)$$
$$\sqrt{x-a} : x \geq a, \quad \frac{1}{x-a} : x \neq a, \quad \log(x-a) : x > a$$

Tricks: In $A - B$, remove from A only. For $g \circ f$, apply f first. Domain traps: root, denominator, log.

4. Groups and Binary Operations

$$* : G \times G \to G, \quad a, b \in G \Rightarrow a * b \in G \text{ closure}$$
$$(a * b) * c = a * (b * c), \quad a * e = e * a = a, \quad a * a^{-1} = a^{-1} * a = e$$
$$a * b = b * a \Rightarrow \text{abelian}$$

Tricks: Identity depends on the operation. Inverse is not always $1/a$. Associativity is grouping; commutativity is order.

5. Complex Numbers

$$z = a + bi, \quad \Re(z) = a, \quad \Im(z) = b$$
$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^n = i^{n \bmod 4}$$
$$\overline{a+bi} = a - bi, \quad |a+bi| = \sqrt{a^2+b^2}, \quad z\bar{z} = |z|^2$$
$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$$
$$z = r(\cos\theta + i\sin\theta), \quad z^n = r^n(\cos n\theta + i\sin n\theta)$$
$$\omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \omega^4 = \omega, \quad \omega^5 = \omega^2$$

Tricks: Reduce powers of i modulo 4; powers of ω modulo 3. Divide complex numbers by multiplying conjugate.

6. Matrices and Determinants

$$A : m \times n, \quad B : n \times p \Rightarrow AB : m \times p$$
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$
$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$
$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0$$
$$\det(AB) = \det A \det B, \quad \det(A^{-1}) = \frac{1}{\det A}, \quad \det(A^T) = \det A$$
$$\det(kA) = k^n \det(A), \quad A \text{ singular} \Rightarrow \det(A) = 0, \quad \operatorname{tr}(A) = \text{diagonal sum}$$

Tricks: Use $ad - bc$, not $bc - ad$. Proportional rows give determinant zero. For AB, use outer dimensions.

7. Quadratic Equations

$$ax^2 + bx + c = 0, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad D = b^2 - 4ac$$
$$D > 0 : \text{real unequal}, \quad D = 0 : \text{real equal}, \quad D < 0 : \text{complex}$$
$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad x^2 - (\alpha+\beta)x + \alpha\beta = 0$$
$$\text{reciprocal roots of } ax^2 + bx + c = 0 : \quad cx^2 + bx + a = 0$$

Tricks: Equal roots means $D = 0$. Sum has minus sign; product does not. Given one root: substitute.

8. Polynomials and Remainder Theorem

$$ax^3 + bx^2 + cx + d = 0$$
$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}$$
$$\text{remainder by } x - a = f(a), \quad f(a) = 0 \Rightarrow (x - a) \text{ factor}$$
$$\text{remainder by } ax + b = f\left(-\frac{b}{a}\right)$$
$$x^3 - y^3 = (x-y)(x^2 + xy + y^2), \quad x^3 + y^3 = (x+y)(x^2 - xy + y^2)$$

Tricks: For $x + 2$, substitute -2. For $2x - 3$, substitute $3/2$. Avoid long division.

9. Partial Fractions

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$
$$\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$
$$\frac{px+q}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

Tricks: Use denominator zeros to isolate coefficients. Repeated factors need each power. Quadratic factor gets numerator $Bx + C$.

10. Sequences and Series

$$a_n = a + (n-1)d, \quad S_n = \frac{n}{2}[2a + (n-1)d], \quad S_n = \frac{n}{2}(a + l)$$
$$a_n = ar^{n-1}, \quad S_n = \frac{a(r^n - 1)}{r - 1}, \quad S_\infty = \frac{a}{1-r}, \ |r| < 1$$
$$AM = \frac{a+b}{2}, \quad GM = \sqrt{ab}, \quad HM = \frac{2ab}{a+b}, \quad AM \geq GM \geq HM$$
$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$
$$\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2, \quad 1 + 3 + \cdots + (2n-1) = n^2$$

Tricks: AP has common difference; GP has common ratio. If a, x, b are in GP, $x^2 = ab$.

11. Permutation and Combination

$$n! = n(n-1)\cdots 1, \quad 0! = 1$$
$${}^nP_r = \frac{n!}{(n-r)!}, \quad {}^nC_r = \frac{n!}{r!(n-r)!}, \quad {}^nC_r = {}^nC_{n-r}$$
$$\text{repeated objects: } \frac{n!}{p!q!r!}, \quad \text{circular arrangements} = (n-1)!$$

Tricks: Arrangement = permutation. Selection = combination. Together = bundle first. Not together = total - together.

12. Probability

$$P(E) = \frac{n(E)}{n(S)}, \quad 0 \leq P(E) \leq 1, \quad P(E') = 1 - P(E)$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$P(A \cap B) = 0 \text{ if mutually exclusive}, \quad P(A \cap B) = P(A)P(B) \text{ if independent}$$
$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(X = r) = {}^nC_r p^r q^{n-r}, \ q = 1 - p$$

Tricks: At least one $= 1 - P(\text{none})$. Dice total 36. Coins total 2^n. Cards total 52. "Or" means union.

13. Binomial Theorem

$$(a+b)^n = \sum_{r=0}^{n} {}^nC_r a^{n-r} b^r, \quad T_{r+1} = {}^nC_r a^{n-r} b^r$$
$$n \text{ even} : T_{n/2+1}, \quad n \text{ odd} : T_{(n+1)/2}, T_{(n+3)/2}$$

Tricks: Coefficient sum: put $x = 1$. Alternating sum: put $x = -1$. Constant term: exponent of x is zero.

14. Trigonometry Basics

$$\sin\theta = \frac{P}{H}, \quad \cos\theta = \frac{B}{H}, \quad \tan\theta = \frac{P}{B}$$
$$\cot\theta = \frac{1}{\tan\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \csc\theta = \frac{1}{\sin\theta}$$
$$\sin^2\theta + \cos^2\theta = 1, \quad 1 + \tan^2\theta = \sec^2\theta, \quad 1 + \cot^2\theta = \csc^2\theta$$
$$\sin 30^\circ = \frac{1}{2}, \quad \sin 45^\circ = \frac{1}{\sqrt{2}}, \quad \sin 60^\circ = \frac{\sqrt{3}}{2}$$
$$\cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}, \quad \cos 60^\circ = \frac{1}{2}$$
$$\tan 30^\circ = \frac{1}{\sqrt{3}}, \quad \tan 45^\circ = 1, \quad \tan 60^\circ = \sqrt{3}$$

Tricks: ASTC. If $\sin\theta = 3/5$, use 3-4-5. If $\tan\theta = 3/4$, then $\sin = 3/5, \cos = 4/5$.

15. Trigonometric Identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$
$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$
$$\sin 2A = 2 \sin A \cos A, \quad \cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$
$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}, \quad \sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}, \quad \cos^2 \frac{A}{2} = \frac{1 + \cos A}{2}$$
$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$
$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$
$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$
$$\max(a \sin x + b \cos x) = \sqrt{a^2 + b^2}$$

Tricks: $\sin 75 \cos 15 - \cos 75 \sin 15 = \sin 60$. Use $\sqrt{a^2 + b^2}$ for max problems.

16. Trigonometric Equations and Graphs

$$\sin x, \cos x : 2\pi, \quad \tan x, \cot x : \pi, \quad \sin(kx), \cos(kx) : \frac{2\pi}{|k|}, \quad \tan(kx) : \frac{\pi}{|k|}$$
$$\sin x = 0 \Rightarrow x = n\pi, \quad \cos x = 0 \Rightarrow x = \frac{\pi}{2} + n\pi, \quad \tan x = 0 \Rightarrow x = n\pi$$
$$-1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1, \quad \tan x \in \mathbb{R}$$

Tricks: Count only in the interval. Period of $\tan(2x)$ is $\pi/2$. If $\sin A = \cos A$ and acute, $A = 45^\circ$.

17. Inverse Trigonometric Functions

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \cos^{-1} x \in [0, \pi], \quad \tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$
$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$
$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x + y}{1 - xy}\right) \text{ with quadrant adjustment}$$

Tricks: Principal branch matters. $\sin^{-1}(\sin x) \neq x$ always.

18. Limits

$$\lim_{x\to 0} \frac{\sin x}{x} = 1, \quad \lim_{x\to 0} \frac{\sin ax}{x} = a, \quad \lim_{x\to 0} \frac{\tan x}{x} = 1$$
$$\lim_{x\to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \quad \lim_{x\to 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x\to 0} \frac{\ln(1 + x)}{x} = 1$$
$$\lim_{x\to a} \frac{x^2 - a^2}{x - a} = 2a, \quad \lim_{x\to a} f(x) = f(a) \Rightarrow \text{continuity at } a$$

Tricks: For 0/0, factor or rationalize. Substitute only after removing zero denominator.

19. Differentiation

$$\frac{d}{dx}c = 0, \quad \frac{d}{dx}x^n = nx^{n-1}, \quad \frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x$$
$$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\cot x = -\csc^2 x, \quad \frac{d}{dx}\sec x = \sec x \tan x, \quad \frac{d}{dx}\csc x = -\csc x \cot x$$
$$\frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}e^{ax} = ae^{ax}, \quad \frac{d}{dx}a^x = a^x \ln a, \quad \frac{d}{dx}\ln x = \frac{1}{x}$$
$$(uv)' = u'v + uv', \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \quad \frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$
$$m = \frac{dy}{dx}, \quad y - y_1 = m(x - x_1), \quad m_n = -\frac{1}{m}$$
$$f'(x) > 0 : \text{increasing}, \quad f'(x) = 0 : \text{stationary}, \quad f''(x) < 0 : \max, \quad f''(x) > 0 : \min$$

Tricks: Derivative means slope. Chain rule is the common trap. For $x^2 \sin x$, use product rule.

20. Integration

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad \int \frac{1}{x}\,dx = \ln|x| + C, \quad \int e^{ax}\,dx = \frac{e^{ax}}{a} + C$$
$$\int \sin ax\,dx = -\frac{\cos ax}{a} + C, \quad \int \cos ax\,dx = \frac{\sin ax}{a} + C$$
$$\int \sec^2 x\,dx = \tan x + C, \quad \int \csc^2 x\,dx = -\cot x + C, \quad \int \sec x \tan x\,dx = \sec x + C$$
$$\int f(g(x))g'(x)\,dx:\ u = g(x), \quad \int_a^b f(x)\,dx = F(b) - F(a), \quad A = \int_a^b y\,dx$$

Tricks: Indefinite integral needs $+C$; definite does not. If derivative of inside is present, use substitution.

21. Analytical Geometry: Straight Lines

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$
$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y = mx + c, \quad y - y_1 = m(x - x_1), \quad \frac{x}{a} + \frac{y}{b} = 1$$
$$Ax + By + C = 0, \quad m_1 = m_2 \text{ parallel}, \quad m_1 m_2 = -1 \text{ perpendicular}$$
$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Tricks: Intercepts a, b give $x/a + y/b = 1$. Vertical slope undefined; horizontal slope zero.

22. Circle

$$(x - h)^2 + (y - k)^2 = r^2, \quad \text{center } (h, k)$$
$$x^2 + y^2 + 2gx + 2fy + c = 0, \quad \text{center } (-g, -f), \quad r = \sqrt{g^2 + f^2 - c}$$
$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0 \text{ diameter form}$$
$$xx_1 + yy_1 = r^2 \text{ tangent to } x^2 + y^2 = r^2$$

Tricks: Complete square. General form uses $2g, 2f$, not g, f.

23. Conic Sections

$$y^2 = 4ax \Rightarrow \text{focus } (a, 0), \quad x = -a \text{ directrix}$$
$$x^2 = 4ay \Rightarrow \text{focus } (0, a), \quad e = 1 \text{ for parabola}$$
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b, \quad \text{major axis} = 2a, \quad c^2 = a^2 - b^2, \quad e = \frac{c}{a}$$
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad \text{transverse axis} = 2a, \quad c^2 = a^2 + b^2, \quad e = \frac{c}{a} > 1$$

Tricks: Parabola eccentricity is 1. Ellipse subtracts: $c^2 = a^2 - b^2$. Hyperbola adds: $c^2 = a^2 + b^2$.

24. Linear Inequalities and Linear Programming

$$ax + by \le c, \quad ax + by = c \text{ boundary line}, \quad Z = ax + by, \quad x \ge 0, y \ge 0 \text{ first quadrant}$$

Tricks: Test $(0, 0)$ to decide side. Maximum/minimum occurs at a vertex. Substitute vertices into Z.

25. Vectors

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \quad |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}, \quad \hat{a} = \frac{\vec{a}}{|\vec{a}|}$$
$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$
$$\vec{a} \cdot \vec{b} = 0 \Rightarrow \text{perpendicular}, \quad |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$$
$$\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}$$
$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

Tricks: Dot zero = perpendicular. Cross zero = parallel. Dot gives scalar; cross gives vector.

26. Three-Dimensional Geometry

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$
$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$$
$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}, \quad l^2 + m^2 + n^2 = 1$$
$$\vec{r} = \vec{a} + \lambda\vec{b}, \quad \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$
$$ax + by + cz = d, \quad \text{normal vector } (a, b, c)$$

Tricks: Direction cosines always satisfy $l^2 + m^2 + n^2 = 1$. Plane coefficients give normal vector.

27. Variation

$$A \propto B \Rightarrow A = kB, \quad A \propto \frac{1}{B} \Rightarrow A = \frac{k}{B}, \quad AB = k$$
$$A \propto BC \Rightarrow A = kBC$$

Tricks: Direct variation: ratio constant. Inverse variation: product constant. Always find k first.

28. Quick MCQ Elimination Rules

- Check sign, domain, quadrant, and positivity before solving.
- Determinant trap: $ad - bc$. Root trap: sum has minus sign.
- Composition order matters. For conics, compare with standard form.
- Probability: distinguish "and" from "or". Arrangement vs selection matters.
- i^n: reduce modulo 4. ω^n: reduce modulo 3.
- Singular matrix: determinant zero. Equal roots: discriminant zero.
- Remainder by $x - a$: put $x = a$. Coefficient sum: put $x = 1$.
- Constant term: exponent of x equals zero. Perpendicular vectors: dot product zero.